

1. Prob 15.13

Mu meson has rest mass of $m_\mu c^2 = 105.7 \text{ MeV}$. If a mu meson is created with a kinetic energy of $T_\mu = 4.1 \text{ MeV}$ during the decay $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$, we have

$$\gamma = 1 + \frac{T_\mu}{m_\mu c^2} \approx 1.0388 \quad \Rightarrow \quad \beta = \sqrt{1 - \frac{1}{\gamma^2}} \approx 0.27 \quad (1)$$

To evaluate the number of quanta per unit energy interval, we can use (15.67)

$$N(\hbar\omega) = \frac{e^2}{\pi\hbar c} \left(\frac{1}{\hbar\omega} \right) \overbrace{\left[\frac{1}{\beta} \ln \left(\frac{1+\beta}{1-\beta} \right) - 2 \right]}^{\approx 0.051} \quad (2)$$

Or, using the approximation that the photons are emitted perpendicular to the direction of the motion of the mu meson, we can evaluate (15.65) at $\theta = \pi/2$ and multiply by $4\pi/\hbar^2\omega$ to obtain

$$N_\perp(\hbar\omega) \approx \frac{e^2}{\pi\hbar c} \left(\frac{1}{\hbar\omega} \right) \beta^2 \quad \text{where } \beta^2 \approx 0.073 \quad (3)$$

We see that the crude $\theta = \pi/2$ approximation overestimates the number of quanta by a factor of ~ 1.5 .

With $\theta = \pi/2$, the maximum photon energy is achieved when the mu meson is at rest. In that case, all 34 MeV of kinetic energy is shared between the photon and the neutrino, which carry equal and opposite transverse momenta. Since both are massless, equal momentum means equal energy, so

$$\hbar\omega_{\max} = 17 \text{ MeV} \quad (4)$$

The number of quanta with energies greater than $\hbar\omega_{\max}/10$ is the integral

$$N_{>\omega_{\max}/10} = \int_{\hbar\omega_{\max}/10}^{\hbar\omega_{\max}} N(\hbar\omega) d(\hbar\omega) = \frac{e^2}{\pi\hbar c} \times 0.051 \int_{\hbar\omega_{\max}/10}^{\hbar\omega_{\max}} \frac{d\omega}{\omega} = 0.051 \times \frac{\alpha}{\pi} \times \ln 10 \approx 2.7 \times 10^{-4} \quad (5)$$

which has the same order of magnitude as the experimental value of $(1.24 \pm 0.25) \times 10^{-4}$.

Note that (15.67) is strictly valid only for soft photons ($\hbar\omega \ll E$), yet here the upper limit $\hbar\omega_{\max} = 17 \text{ MeV}$ exceeds the mu meson kinetic energy of 4.1 MeV. However the integral is logarithmic and dominated by the low- ω end where the soft-photon approximation holds. The overestimate at the upper end partly explains why our result exceeds the experimental value.

2. Prob 15.14

In internal conversion, an electron is suddenly ejected with kinetic energy $T_e = 1 \text{ MeV}$. The electron rest mass is $m_e c^2 = 0.511 \text{ MeV}$, so

$$\gamma = 1 + \frac{T_e}{m_e c^2} \approx 2.96 \quad \Rightarrow \quad \beta = \sqrt{1 - \frac{1}{\gamma^2}} \approx 0.94 \quad (6)$$

From (15.67), the number of quanta emitted per unit energy interval is

$$N(\hbar\omega) = \frac{\alpha}{\pi} \left(\frac{1}{\hbar\omega} \right) \overbrace{\left[\frac{1}{\beta} \ln \left(\frac{1+\beta}{1-\beta} \right) - 2 \right]}^{\approx 1.71} \quad (7)$$

The maximum photon energy is $\hbar\omega_{\max} \approx T_e = 1 \text{ MeV}$ (the electron can radiate essentially all of its kinetic energy).

For each internal conversion, we calculate the number of photons carrying away more than $\Delta = 0.01 T_e$ as

$$N_{>\Delta} = \int_{\Delta}^{\hbar\omega_{\max}} N(\hbar\omega) d(\hbar\omega) \approx 1.71 \times \frac{\alpha}{\pi} \times \ln 100 \approx 0.018 \quad (8)$$

With this small number, we can ignore the possibility of multiple photon emissions. Then it follows that it takes $1/N_{>\Delta}$ electron ejections to produce one photon with energy greater than Δ , hence the fraction of electrons having energies less than $0.99 T_e$ is just $N_{>\Delta} = 1.8\%$. This level of low-energy tail is readily observable experimentally.

As in problem 15.13, the soft-photon approximation is approximately applicable because of the logarithmic nature of the integral.

3. Prob 15.15

(a) Let $\mathbf{p}_{+1}, \mathbf{p}_{+2}, \mathbf{p}_-$ be the momenta of the resulting pions. Conservation of momentum requires

$$\mathbf{p}_{+1} + \mathbf{p}_{+2} + \mathbf{p}_- = 0 \quad \Rightarrow \quad \boldsymbol{\beta}_{+1} + \boldsymbol{\beta}_{+2} = -\boldsymbol{\beta}_- \quad (9)$$

On the other hand, for nonrelativistic pions, (15.9) gives

$$\begin{aligned} \lim_{\omega \rightarrow 0} \frac{d^2 I}{d\omega d\Omega} &\approx \frac{e^2}{4\pi^2 c} |\mathbf{n} \times [\mathbf{n} \times (\boldsymbol{\beta}_{+1} + \boldsymbol{\beta}_{+2} - \boldsymbol{\beta}_-)]|^2 \\ &= \frac{e^2}{\pi^2 c} |\mathbf{n} \times (\mathbf{n} \times \boldsymbol{\beta}_-)|^2 = \frac{e^2}{\pi^2 c} \beta_-^2 \sin^2 \theta = \frac{2e^2}{\pi^2 c} \frac{T_-}{m_\pi c^2} \sin^2 \theta \end{aligned} \quad (10)$$

(b) Integrating part (2) over solid angles

$$\frac{dI}{d\omega} = \int \frac{d^2 I}{d\omega d\Omega} d\Omega = \frac{16e^2}{3\pi c} \frac{T_-}{m_\pi c^2} \quad (11)$$

Giving the number of photons per unit energy

$$N(\hbar\omega) = \frac{1}{\hbar^2 \omega} \frac{dI}{d\omega} = \frac{16\alpha}{3\pi} \frac{T_-}{m_\pi c^2} \frac{1}{\hbar\omega} \quad (12)$$

The branching ratio for emission of a photon with energy $\hbar\omega > \Delta$ is obtained by integrating up to a cutoff $\hbar\omega_{\max} = 75\text{MeV}$ (the total energy release) and averaging T_- over all allowed energy splits among the three pions. For three equal-mass nonrelativistic particles all splits are equally likely, so $\langle T_- \rangle = \hbar\omega_{\max}/3 = 25\text{MeV}$, and $m_\pi c^2 \approx 140\text{MeV}$

$$\begin{aligned} N_{>\Delta} &= \int_{\Delta}^{\hbar\omega_{\max}} \frac{16\alpha}{3\pi} \frac{\langle T_- \rangle}{m_\pi c^2} \frac{d\omega}{\omega} = \frac{16\alpha}{3\pi} \frac{\langle T_- \rangle}{m_\pi c^2} \ln\left(\frac{\hbar\omega_{\max}}{\Delta}\right) \approx 0.0022 \times \ln\left(\frac{75 \text{ MeV}}{\Delta}\right) \\ &= \begin{cases} 9.6 \times 10^{-3} & \Delta = 1\text{MeV} \\ 4.5 \times 10^{-3} & \Delta = 10\text{MeV} \\ 4.2 \times 10^{-3} & \Delta = 11\text{MeV} \end{cases} \end{aligned} \quad (13)$$

agreeing with experimental values qualitatively.

4. Prob 15.16

(a) The radiation comes from the sudden acceleration of π^+ , its distribution is given by (15.65)

$$\frac{d^2 I}{d\omega d\Omega} = \frac{e^2}{4\pi^2 c} \frac{\beta^2 \sin^2 \theta}{(1 - \beta \cos \theta)^2} \quad (14)$$

where θ is the angle between the photon and the π^+ direction.

It is elementary to show that at $\cos \theta = \beta$, (14) reaches its maximum value. So $\cos \theta = \beta$ is the most likely direction for the photons to be emitted. For $\beta = 0.71$, the most likely angle is $\theta \approx 45^\circ$. This agrees with figure 6 of Edwards et al. which shows the distribution of events happen within the forward hemisphere along the π^+ direction.

(b) Let's first calculate the nonradiative kinetic energy of the π^+ . For the two-body decay $K^+ \rightarrow \pi^+ \pi^0$ in the K rest frame, let the resulting particles have energy E_{π^+}, E_{π^0} and momenta $\mathbf{p}_{\pi^+}, \mathbf{p}_{\pi^0}$, so that

$$\mathbf{p}_{\pi^+} + \mathbf{p}_{\pi^0} = 0 \quad E_{\pi^+} + E_{\pi^0} = M_K c^2 \quad (15)$$

Also from

$$E_{\pi^+}^2 = p_{\pi^+}^2 c^2 + M_{\pi^+}^2 c^4 \quad E_{\pi^0}^2 = p_{\pi^0}^2 c^2 + M_{\pi^0}^2 c^4 \quad (16)$$

we have

$$E_{\pi^+}^2 - E_{\pi^0}^2 = (M_{\pi^+}^2 - M_{\pi^0}^2) c^4 \quad \Rightarrow \quad E_{\pi^+} = \left(\frac{M_K^2 + M_{\pi^+}^2 - M_{\pi^0}^2}{2M_K} \right) c^2 \approx 248.1\text{MeV} \quad (17)$$

giving the nonradiative kinetic energy of the π^+ as

$$T_{\pi^+}^{NR} = E_{\pi^+} - M_{\pi^+}c^2 \approx 108.5\text{MeV} \quad (18)$$

as well as

$$\gamma_0 = \frac{E_{\pi^+}}{M_{\pi^+}c^2} \approx 1.77 \quad \Rightarrow \quad \beta_0 = \sqrt{1 - \frac{1}{\gamma_0^2}} \approx 0.83 \quad (19)$$

In the idealization where π^0 carries its nonradiative momentum and the kinetic energy of π^+ is in the range $55\text{MeV} \leq T_{\pi^+} \leq 90\text{MeV}$, the energy of the radiated photon must be in the range

$$\hbar\omega_{\min} = T_{\pi^+}^{NR} - 90\text{MeV} = 18.5\text{MeV} \quad \hbar\omega_{\max} = T_{\pi^+}^{NR} - 55\text{MeV} = 53.5\text{MeV} \quad (20)$$

In the soft-photon limit ($\beta = \beta_0 = 0.83$), we integrate (15.67) over the photon energy range

$$N = \int_{\hbar\omega_{\min}}^{\hbar\omega_{\max}} \frac{\alpha}{\pi} \left(\frac{1}{\hbar\omega} \right) \overbrace{\left[\frac{1}{\beta} \ln \left(\frac{1+\beta}{1-\beta} \right) - 2 \right]}^{\approx 0.85} d(\hbar\omega) \approx 0.85 \times \frac{\alpha}{\pi} \times \ln \left(\frac{53.5}{18.5} \right) \approx 2.1 \times 10^{-3} \quad (21)$$

This is the probability of emitting a photon in this energy range per $K^+ \rightarrow \pi^+\pi^0$ decay. Since the $\pi^+\pi^0$ mode is 21% of all K^+ decays, the overall branching ratio is thus

$$0.21 \times 2.1 \times 10^{-3} \approx 4.4 \times 10^{-4} \quad (22)$$

This is within a factor of ~ 1.6 of the experimental value $(2.75 \pm 0.15) \times 10^{-4}$. The overestimate is the result of using soft-photon approximation at the upper end of the range.